

Real-time precision crop identification in high weed-density environments for robotic weed control using spectral fluorescence imaging in celery

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ABSTRACT

This paper presented an a pioneering method for real-time robotic weed control in vegetable fields characterized by high weed densities, particularly in challenging scenarios where the foliage of crop plants was intricately intertwines with weeds, causing significant occlusion. This issue was particularly pronounced in organic farming environments where conventional classification algorithms often fell short. Our proposed method, termed *crop signalling*, involved the pre-transplantation treatment of celery crop plants with Rhodamine B (Rh-B), a fluorescent compound with unique optical properties that generated distinct signals readable by machines to effectively discern crop plants from weeds. A custom illumination system was specifically designed to excite the fluorescence properties of the Rh-B dye, facilitating visualization. Concurrently, a dedicated machine vision algorithm was developed not only to differentiate crop plants from weeds but also to accurately determine the stem locations of crop plants as they enter the soil, facilitating targeted weed-knife control. Experimental results showcased exceptional performance, achieving a 100% accuracy rate in detecting and distinguishing crop plants from weeds in densely populated fields, with no instances of false positives. The algorithm demonstrated a high precision rate of 99.66% in identifying celery plant stem locations across 313 images. This research represented a significant advancement in spectral fluorescence imaging for precision weed and crop classification, offering promising prospects for the application of robotic weed control in celery cultivation.

1. Introduction

Weed management in agricultural fields is a persistent challenge that directly impacts crop yield, quality, and overall profitability. Traditional approaches, including manual weeding, herbicide application, and mechanical methods, became increasingly inadequate and labor intensive, particularly in organic farming systems where the use of synthetic herbicides is restricted (Fennimore et al., 2019; Fennimore et al., 2016). Organic vegetable farming, such as celery cultivation, posed additional complexity due to the dense growth of weeds, which often intertwined with crop foliage, causing occlusion and making mechanical weed control difficult without risking damaging the crops (Slaughter, 2008). These challenges highlighted the need for innovative, sustainable, and efficient weed control strategies.

Recent advancements in artificial intelligence (AI), machine learning

(ML), and robotics have significantly revolutionized precision agriculture, addressing challenges such as crop-weed discrimination. Machine learning techniques (Lottes et al., 2018b; Knoll et al., 2019; Islam et al., 2021) and deep learning (DL) methods (Lottes et al., 2018a; Hu et al., 2024; Jin et al., 2022) proved effective in distinguishing crops from weeds, enhancing precision while reducing labor demands. For instance, Lottes and Stachniss (2017) introduced a semi-supervised online classification method leveraged plant arrangement for accurate crop-weed classification. Similarly, Su et al. (2021) and You and Lee (2020) utilized data augmentation and semantic segmentation to enhance classification accuracy.

Addressing occlusion in overlapping plants, Zhang et al. (2022) applied support vector machines (SVM) with image blocking techniques, offered progress but lacked the precision required for real-time use in densely planted fields. Zhao and Zhai (2022) developed a machine learning system for cabbage and weed identification, integrating a target

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Nomenclature

$I_o(x, y)$	represents the original image captured under green light illumination
T_{low}, T_{high}	defined thresholds with low and high intensity values
$I_t(x, y)$	the segmented binary image
$I_{eroded}(x, y), I_{dilated}(x, y)$	the eroded and dilated binary image, respectively
$(x_a, y_a), (x_d, y_d)$	actual and desired soil-entry location
n	number of images
t	represents t^{th} image
CNNs	convolution neural networks
FOV	field of view
AUC	under the Curve
LEDs	light emitting diodes
UV	ultraviolet
NIR	near-infrared
Vis	virtual instruments
RMSE	root-mean-square error

spraying system. While effective in lower-density scenarios, this approach struggled in dense, intertwined crop environments, unlike our Rh-B-based method.

Algorithmic advancements such as hybrid approaches combining deep neural networks (DNNs) with SVMs were explored (Julie et al., 2021; Tao and Wei, 2022; Kempegowda et al., 2022; Espejo-Garcia et al., 2020), improving classification accuracy. However, these methods lacked crop-specific signaling mechanisms crucial for challenging environments. Agriculture 5.0 technologies and their transformative potential were reviewed by Juwono et al. (2023), while Adhinata et al. (2024) identified gaps in ML and DL applications for high weed densities and complex occlusions. Transfer learning was proposed to improve weed identification across diverse environments (Espejo-Garcia et al., 2020), but its practical application remained limited, particularly in organic farming with intricate weed-crop interactions.

Spectral imaging emerged as a promising approach for crop-weed differentiation. Slaughter et al. (2004, 2008) demonstrated the potential of hyperspectral models for distinguishing crop plants, achieving accuracies up to 95 % in controlled settings. However, challenges such as overlapping foliage and environmental variability often degraded classification accuracy (Sa et al., 2018). Recent efforts have introduced deep neural networks (DNNs) and hybrid methods, such as combining DNNs with support vector machines (SVMs), for weed detection (Julie et al., 2021; Kempegowda et al., 2022). While these methods enhanced classification accuracy, their effectiveness in real-time, high weed-density, occluded environments remained limited.

The innovative concept of crop signaling offered a promising avenue for addressing these challenges. Raja et al. (2019) introduced crop signaling compounds to enhance crop visibility in automated systems, achieving classification accuracies of over 99 % in tomato and lettuce fields (Raja et al., 2020a, Raja et al., 2020b). These methods, incorporating systemic and plant-level marking techniques, demonstrated real-time reliability. Despite its success, crop signaling had yet to be optimized for celery, where the challenges of intertwined weeds and crops persisted.

Building on this foundation, this study focused on using Rhodamine B (Rh-B) as a pre-transplant signaling compound for celery. The novel application of Rh-B bridged the gap between crop signaling research and robotic weed control systems, addressing the unmet need for effective weed management in dense, intertwined conditions. By optimizing Rh-B dosage for precise fluorescence imaging, this research overcame occlusion challenges and ensured crop safety. The integration of fluorescence imaging with advanced computer vision algorithms enabled the precise

identification of celery stems, facilitating targeted robotic interventions. Unlike conventional RGB imaging, the proposed system leveraged multi-wavelength fluorescence data to accurately locate the soil-entry point of celery stems, enabling targeted robotic interventions.

The contributions of this study were threefold:

- Developed a sustainable crop signaling method using Rh-B for celery, optimized for high-precision fluorescence imaging.
- Designed a robust computer vision algorithm capable of differentiating crops from intertwined weeds in real-time.
- Integrated and validated the proposed solution in a robotic weed control system, addressing the unique challenges of high weed density of organic celery farming.

In terms of system design, this study pioneered the integration of a photometric camera with specialized band-pass filters and controlled illumination to overcome field-specific challenges such as occlusion and variability in weed density. The system was optimized for real-time performance, demonstrated through rigorous testing in simulated environments, and showed promise for autonomous field deployment. These innovations collectively represented a sustainable and efficient solution for celery farming, particularly in organic agriculture, where traditional methods often struggle.

2. Materials and methods

This section introduced a novel crop signaling technique employing Rh-B dye, with the purpose of generating distinct visual features for crop plants. The crop signaling or marking process was described in Section 2.1. We developed a specialized machine vision system which was presented in Section 2.2, tailored for capturing outdoor images in commercial vegetable fields, which was specifically designed to focus on crop plants marked with a unique crop signaling compound to unequivocally differentiate from weeds, even in the presence of a very high weed densities. The overarching goal of this machine vision system was to formulate an algorithm dedicated to the identification of crop plants to distinguish them from surrounding weeds in a high weed density environments, as described in Section 2.3. This algorithm was instrumental in creating a precise plant mapping, which, in turn, served as the foundational basis for the implementation of targeted herbicide spraying.

2.1. Crop signaling and marking technique

Crop signaling represented a groundbreaking method aimed at reliably differentiating crop plants from weeds, crucial for the implementation of an autonomous weed control system. In this approach, a unique compound was applied to crop plants before planting, generating an optical signal detectable by machines. Subsequently, during the weed control phase, a weeding robot utilized this crop signal to classify and precisely locate the position of crop plants. For this study, we adopted a strategic crop signaling technique, treating seedlings with a signaling compound before transplanting. The development of this system necessitated considerations for its robustness in the natural outdoor environment of a commercial farm. The applied compound quantity was limited to avoid impacting crop growth, yet sufficient for real-time signal detection. The compound's longevity, regardless of environmental conditions, was considered crucial for effective weed control throughout the season. The viability of the crop signaling technique relied on successful compound selection that does not adversely affect crops or the environment, ensuring a readable signal for machine classification. Cost-effectiveness was also considered a key criterion for on-farm implementation.

Recognizing the promising application of fluorescent markers in various domains (Harper et al., 1999; Byrd et al., 2002) including plant research (Chin et al., 2018), our study identified a fluorescent marker as

the suitable compound for marking crop plants. This fluorescent marker generated a unique machine-readable signal utilized for robotic weed control. Experimental results underscored the material's potential for crop detection, exhibiting no false positive errors in the absence of natural compounds with autofluorescence at similar wavelengths as the signaling compound.

In this research, Rhodamine B (Rh-B), a fluorescent marker, was employed for marking celery seedlings. Rh-B ([9-(2-carboxyphenyl)-6-diethylamino-3-xanthenylidene]-diethylammonium chloride, $\geq 99\%$, analytical grade, Sigma-Aldrich, St. Louis, MO, USA) adhered to the regulations outlined in 40 CFR 180.2020 and was listed on the U.S. Environmental Protection Agency (EPA) 4B list. This designation indicated that the EPA possessed adequate information to reasonably conclude that the current utilization of Rh-B will not have adverse effects on public health or the environment. As per section 180.2020 of 40 CFR Ch. I (7–1–07 Edition, the Code of Federal Regulations), the use of Rh-B (not exceeding 2 % by weight of the formulated product and 60 ppm on the treated seed) did not necessitate a tolerance or exemption, based on the EPA's determination that it was unlikely to result in residues on or in food.

A total of 313 celery seedlings (origin: USA, variety: Versatile) were exposed to a 60 ppm Rh-B aqueous solution. The celery roots were immersed in the Rh-B solution for 48 h. After treatment, the celery plants were transferred to plastic pots containing moist soil and were grown outdoors in Salinas, CA, USA, under full sunlight for a period of 27 days. The outdoor environment exhibited a daily temperature range of 11.7 to 26.1 °C, relative air humidity ranging from 57 to 100 %, and solar radiation fluctuating between 515 to 660 W/m². More about the treatment process and the results could be found in [Su et al., 2020](#). Simultaneously, mustard and dill plants were selected to serve as proxies for broadleaf weed species. These plants were chosen due to their similar morphological traits (such as leaf shape and structure), which are commonly encountered in celery field and typically overshadow crops like celery. The goal was to create a controlled experimental environment using readily available crops with similar leaf structures. These plants were grown in a greenhouse, received daily watering. On day 28 (July 14), the celery crop plants, along with mustard and dill, were

imaged using an acquisition system for subsequent analysis.

2.2. Fluorescence macroscopic image acquisition system

A fluorescence macroscopic image acquisition system was developed at the University of California, Davis, to capture high-quality images of celery crop fields. The imaging system was designed with careful optimization of key parameters to ensure robust performance under field-representative conditions. The system utilized a high-resolution photometric monochrome camera (model: Prime 95B, Tucson, AZ, USA) with a 25 mm lens and a Nikon 50 mm f/1.8D lens mounted on top of a closed imaging chamber. The chamber (dimensions: 0.45 m × 0.45 m × 1.8 m) was constructed with black vinyl sheets on all sides to block ambient light as shown in [Fig. 1](#). A mechanical skid minimized sunlight leakage from the edges along the soil surface. This setup ensured consistent illumination, which was critical for isolating fluorescence signals from Rhodamine B (Rh-B) under varying external conditions.

2.2.1. Optimization of camera and illumination parameters

The camera's intrinsic parameters—focal length, focus, aperture, exposure time, and gain—were systematically optimized through a combination of theoretical and experimental approaches:

- **Focus and Aperture:** The aperture was adjusted based on the height of the plants to achieve an optimal depth of field. For larger plants (>12 in.), an aperture of f/5.6 or higher was used, while smaller plants required a lower f-stop (e.g., f/2.8) to maximize light capture and shorten exposure time. These settings were fine-tuned through iterative testing to ensure sharpness across the plant height.
- **Gain and Exposure Time:** Gain adjustments were performed using LabVIEW software in combined gain mode to balance noise and brightness. Exposure times were calibrated so that the brightest pixels reached 80–90 % saturation, preventing overexposure while maximizing signal clarity.
- **Lighting Optimization:** The system employed 12 green LEDs (Model: LZ4-40G108-0000, peak emission: 523 nm) with sharpening filters (cut-off wavelength: 550 nm). Adjustable power supplies controlled

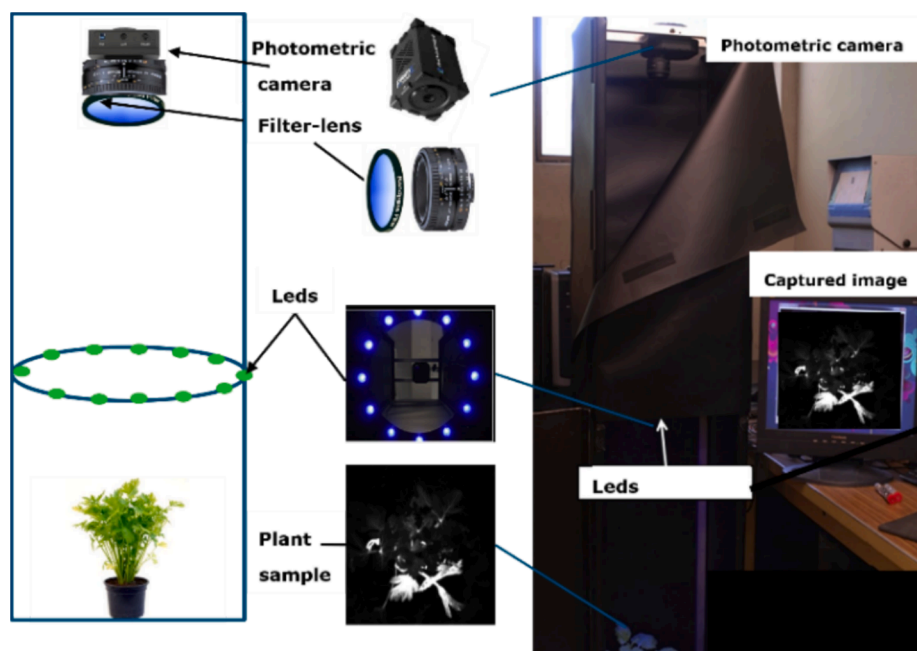


Fig. 1. Experimental setup of the fluorescence macroscopic imaging systems equipped with (a) an optical bandpass filter (Semrock: FF03-575/25–50.8-D 575/25), (b) a Nikon lens (AF Nikkor 50 mm f/1.8D), (c) a photometric camera (model: prime 95B 25 mm, 3440 East Britannia Drive, Tucson, AZ, USA), (d) 12 green light emitting diodes.

LED brightness, ensuring that fluorescence signals from the Rh-B compound did not saturate the camera.

2.2.2. Handling filter-induced variations

The use of an optical band-pass filter (Semrock FF03-575/25–50.8-D) introduced slight focus variations. To address this, a smaller aperture was used to increase the depth of field, thereby ensuring consistent focus across all experimental conditions.

2.2.3. Controlled imaging and image processing

Images were acquired while the imaging chamber was mounted on a conveyor system simulating commercial tractor speeds (2 km/h). The camera was connected via USB 3.0 to an onboard computer (Intel Core i7, 3.4 GHz, with 12 GB RAM) for acquiring images shown in Fig. 2. LabVIEW software was used to control the photometric camera, capturing sequences of high-resolution images for subsequent analysis. Open-source image analysis software (Fiji ImageJ) was employed to measure hue, saturation, and value (HSV) parameters. These measurements facilitated precise discrimination between Rh-B-treated celery plants and surrounding weeds.

2.2.4. Validation and performance testing

The imaging system was rigorously tested to ensure robust performance:

- **Brightness Uniformity:** The system achieved consistent fluorescence signal detection across a range of sample densities.
- **Signal-to-Noise Ratio (SNR):** Optimized camera settings yielded high SNR values, enabling reliable signal extraction even in challenging conditions.
- **Conveyor Speed Testing:** The system maintained 100 % classification accuracy at a conveyor speed of 2 km/h, demonstrating its suitability for real-time applications.

By integrating an optimized imaging system with precise control over camera and illumination parameters, this setup represents a scalable solution for fluorescence imaging in agricultural robotics. The methodology not only ensures robust signal acquisition but also forms the foundation for the subsequent development of classification algorithms and field validation.

2.3. Experimental setup

To simulate in-row weed control in a celery field under controlled indoor conditions, celery plants treated with Rhodamine B (a crop signalling compound) were transplanted into pots measuring 50.8 mm × 50.8 mm. The celery plants varied in size, with heights ranging from 100 to 150 mm and widths from 60 to 170 mm. Mustard and dill plants were used to represent weeds, grown in flats with individual cells of 25.4 mm × 25.4 mm. Weed dimensions spanned from 50 mm to 150 mm in height and 10 mm to 120 mm in width.

The experiments were conducted four weeks after planting, when the plants reached a height of 100–200 mm, following standard commercial farming practices, including drip irrigation. In outdoor celery fields, in-row weed control typically involves a tractor equipped with an imaging system to classify crops and weeds, and an actuator to remove weeds. To replicate these conditions indoors, an experimental setup was established at the University of California, Davis, as shown in Fig. 3.

This setup comprised a conveyor belt and pulley arrangement to simulate tractor movement, an imaging system for real-time plant classification, and an automatic knife mechanism for weed removal.

2.3.1. Mechanical setup

The conveyor belt system consisted of two pulleys, with one powered by an adjustable AC driver (Lenze AC Tech SMVector, USA). Seven aluminum pots (178 mm × 280 mm) were affixed sequentially to the belt, each accommodating flats containing celery plants positioned centrally, surrounded by mustard and dill as weeds, as illustrated in Fig. 4. The celery plants varied in size, with heights ranging from 100 to 150 mm and widths from 60 to 170 mm. Weed dimensions spanned from 50 mm to 150 mm in height and 10 mm to 120 mm in width. Celery crop plants were consistently positioned at a fixed distance, while weeds were randomly placed around them. Weed density was categorized as follows: light density, representing 30 % or less coverage of the image area by weeds; moderate density, indicating 30 % to 70 % coverage; and heavy density, signifying over 70 % coverage. The coverage area calculations excluded the regions covered by the crop plants.

2.3.2. Imaging and knife mechanism

The imaging system used a photometric camera to capture images of the plants as they moved through the conveyor. These images were

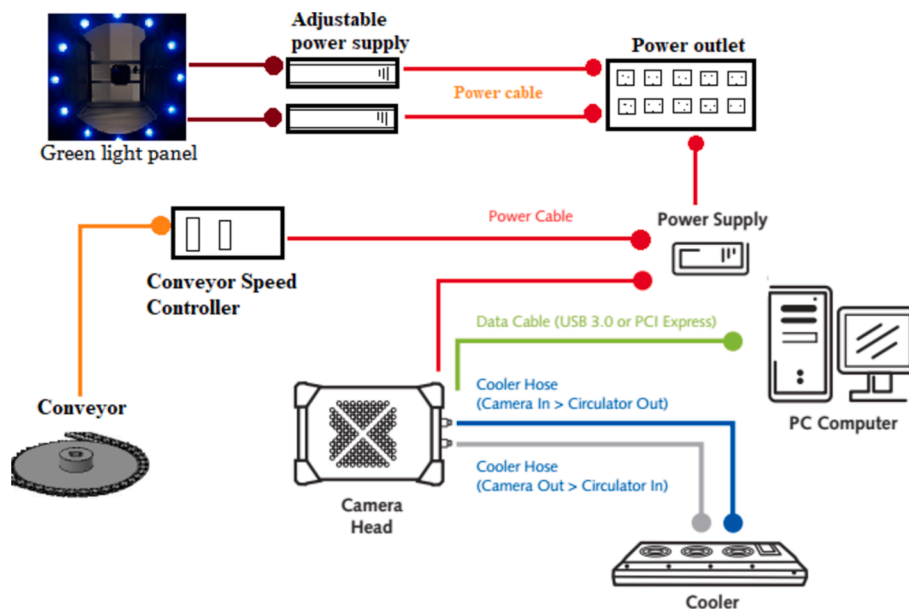


Fig. 2. Connectivity diagram of camera, LEDs and conveyors setup to conduct the experiments for fluorescence image acquisition and classification of crop plants from weeds.

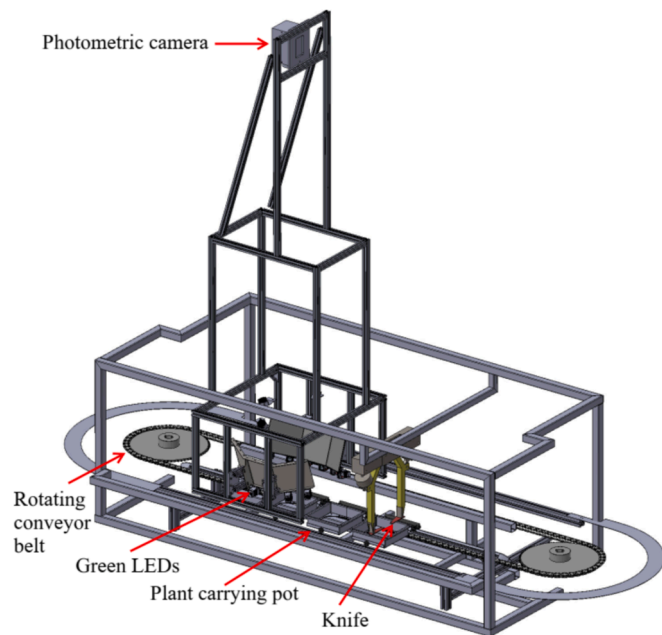


Fig. 3. The mechanical structure to conduct experiments for automatic weed control in indoor environment.

analyzed to identify crop plants and estimate their locations relative to the knife mechanism. Positioned behind the imaging chamber, the knife mechanism consisted of two triangular blades made from hardened tool steel (Model Aristocrat D-2, USA). The blades were mounted on a linkage system that allowed precise control of their movement, enabling effective weed removal while safeguarding crop plants. The knife mechanism was actuated by a solenoid air control valve (Model A212SD024-D, Ingersoll Rand, Ireland), which controlled the pneumatic cylinders. When a crop plant approached, the knife remained open, closing only when the crop passed a predefined safety zone, as shown in

Fig. 5.

This ensured accurate weed removal without damaging the celery plants.

2.3.3. Simulated weed control operations

The system operated iteratively, capturing images, analyzing them to identify crops and weeds, and actuating the knife for weed removal. Each cycle continued until the end of the simulated row. Fig. 4a provided a side view of the conveyor belt system, showing the sequential arrangement of pots, while Fig. 4b illustrated the spatial configuration of celery plants surrounded by weeds in a single tray.

This experimental setup effectively replicated the spatial and operational conditions encountered in real agricultural fields, enabling realistic assessments of the automated weed control system's performance. The setup ensured precision weed removal, maintaining a weed-free environment around the celery plants and validating the feasibility of the proposed system for in-field applications.

3. Crop detection and classification algorithm

This study introduced a novel in-row crop detection and classification algorithm tailored to facilitate automated weed control using a robotic knife. The algorithm leveraged a crop signaling technique combined with machine vision to operate effectively in dense weed environments where crop and weed foliage frequently overlap. Traditional classification methods often faced challenges in such scenarios due to variability in weed shapes, sizes, and occlusion issues.

The proposed system employed Rhodamine B (Rh-B) as a fluorescent marker for crops, enabling precise differentiation between crops and weeds under controlled green light illumination. A monochrome photometric camera captured images, which were processed in real time to identify and localize crop plants to distinguish them from weeds. The novel contribution of this work lies in its ability to robustly classify crops amidst high weed density and accurately determine the soil-entry location of crop stems for robotic knife guidance.

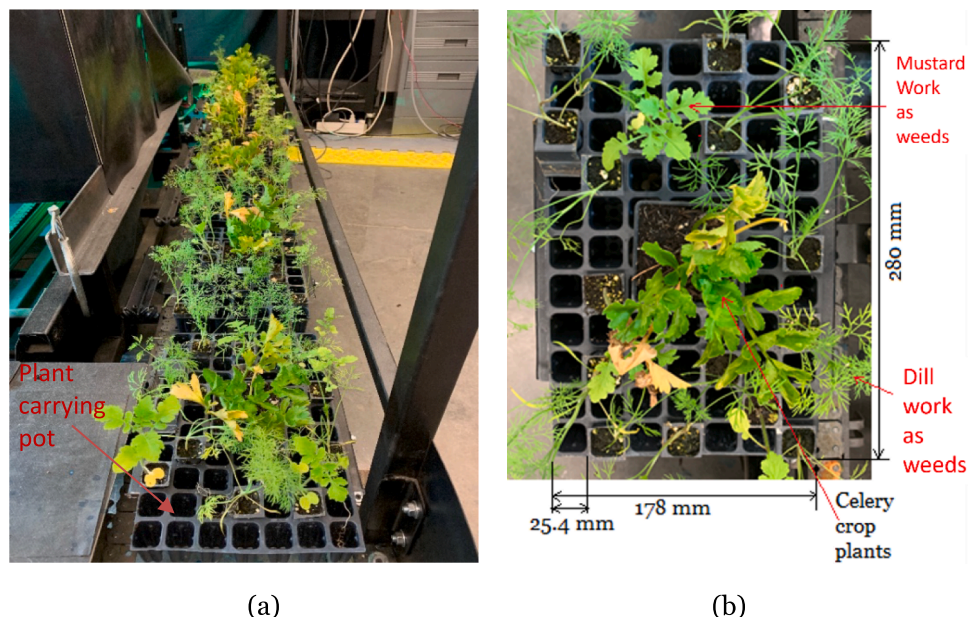


Fig. 4. Illustrates the simulated in-row-celery crop scenario with densely placed weeds within plant-carrying pots on a conveyor belt. (a) the perspective provided a side view of the conveyor belt system, depicting the sequential arrangement of pots carrying celery crop plants along with densely positioned weeds. (b) offered a top-down view of a tray carrying celery crop plants positioned centrally, surrounded by randomly placed mustard and dill as weeds. The dimensions of the tray were also presented, providing clarity on the spatial layout and configuration of the simulated crop arrangement. The conveyor moved at a defined speed of 2.4 km/h, simulating the traversal of a tractor in an outdoor field.

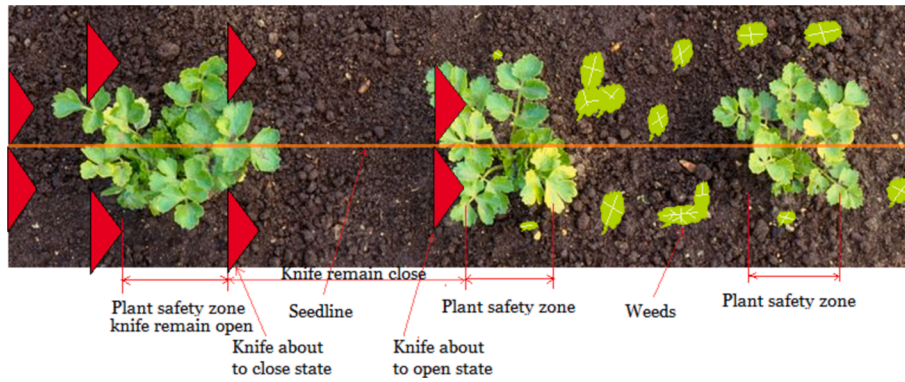


Fig. 5. A concept showing a knife cutting weeds near celery plants with the two operating states (open and close) of the knife.

3.1. Image processing workflow

The imaging system operated at a conveyor speed of 2.4 km/h, surpassing the speeds achieved by prior methods. This high-speed operation was made possible by simplifying the crop detection process through the crop signaling concept and integrating optimized image processing techniques, as presented in Fig. 6.

3.1.1. Image capture and preprocessing

The images were acquired within an indoor experimental setup utilizing the system depicted in Fig. 3, with the conveyor belt in motion at a speed of 2.4 km h⁻¹. In Fig. 7a, the captured image by the monochrome photometric camera is presented, taken under green light illumination. The raw images captured, shown in Fig. 7a were represented as $I_o(x, y)$, where x and y denoted pixel coordinates. An intensity-based thresholding technique was applied to segment the crop signal as follows:

$$I_t(x, y) = \begin{cases} 255 & \text{if } T_{low} \leq I_o(x, y) \leq T_{high} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Here, $T_{low} = 108$, $T_{high} = 189$ are empirically determined thresholds that separate the crop signal from weeds and background. This original raw image was analyzed using ImageJ to show to the readers that the celery crop plants and weeds foliage's are intermingled with each other, revealing the difficulty in analyzing to clearly distinguish between the crop and weeds, as depicted in Fig. 7b.

3.1.2. Noise reduction and object isolation

Connected component analysis is performed on the binary image shown in Fig. 8a, represented as $I_t(x, y)$ to remove noise. Components with an area smaller than $P = 100$ pixels are discarded to obtained eroded binary image $I_{eroded}(x, y)$ shown in Fig. 8b as,

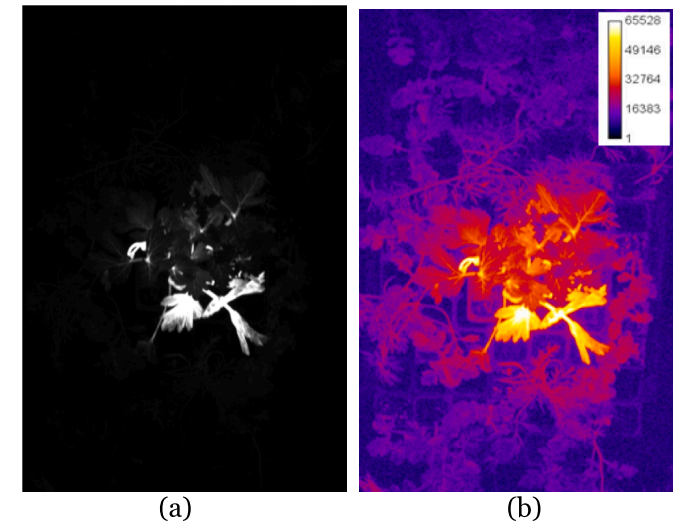


Fig. 7. (a) captured original monochrome image by the camera under green light illumination, (b) analyzed original image using ImageJ to show celery crop plants and randomly placed weeds where weeds and crop foliage's are intermingled with each other.

$$I_{eroded}(x, y) = \text{Erosion}(I_t(x, y)) \quad (2)$$

Morphological dilation operation was then executed on $I_{eroded}(x, y)$ to gradually enlarge the boundaries of regions of foreground pixels (i.e. white pixels) in the images. Thus, areas of foreground pixels grow while holes within those regions become smaller. The dilated binary image $I_{dilated}(x, y)$, was obtained, as shown in Fig. 8c, using:

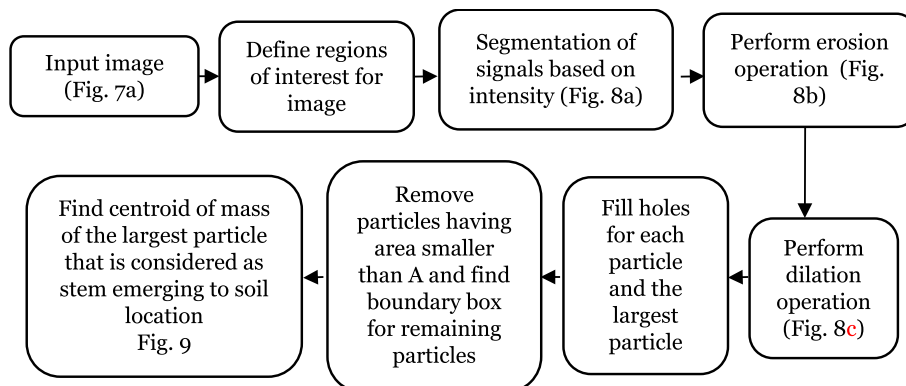


Fig. 6. Flow chart of the proposed machine vision algorithm for classification of weed and crop.

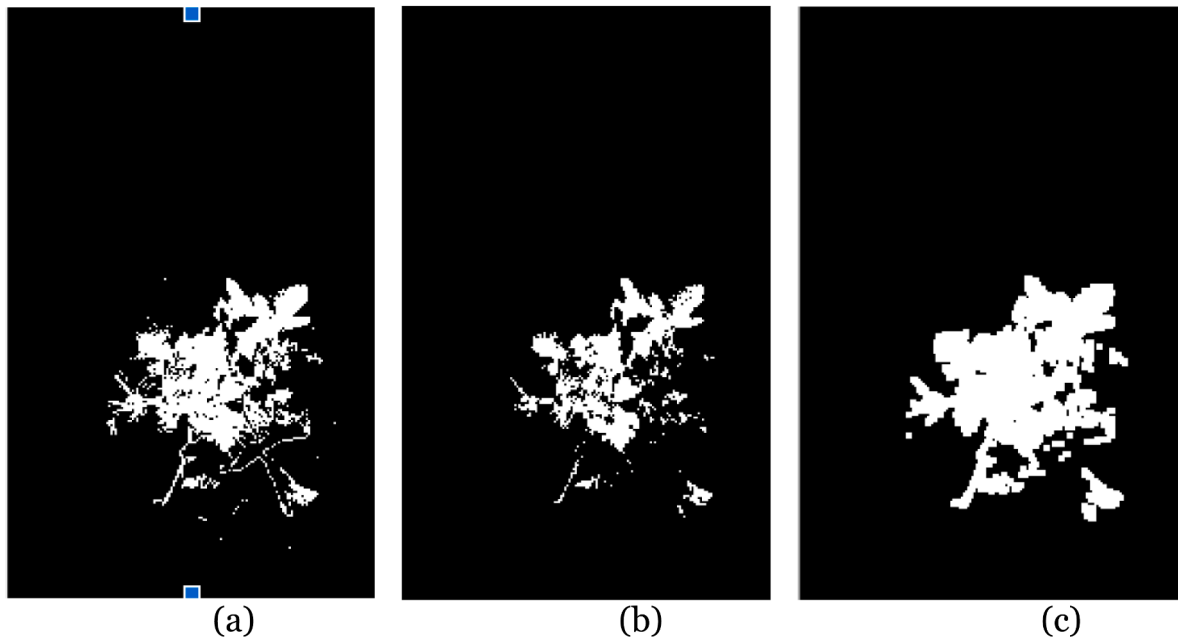


Fig. 8. (a) segmented binary image, (b) image after erosion operation, refining boundaries and reducing noise, (c) image after dilation operation, restoring objects size and shape.

$$I_{dilation}(x,y) = \text{Dilation}(I_{eroded}(x,y)) \quad (3)$$

Subsequently, a morphological fill-holes operation was performed on $I_{dilated}(x,y)$ dilated binary image to get the continuous leaf/stem areas of crop plants. It changes connected background pixels (black pixels) to foreground pixels (white pixels), stopping when it reaches object boundaries. The final result, $I_{filled}(x,y)$, was achieved as follows:

$$I_{filled}(x,y) = \text{FillHoles}(I_{dilation}(x,y)) \quad (4)$$

3.1.3. Crop identification

A particle analysis was performed to remove small particles having areas less than A pixels² and to identify the boundary boxes (green) for remaining objects in the image shown in Fig. 9, which were considered

as crop plants (red). Objects representing crops were enclosed in bounding boxes, and their centroids were calculated to determine the soil-entry location of crop stems:

$$(x_a, y_a) = \frac{1}{n} \sum_{i=1}^n (x_i, y_i), \quad (5)$$

where n was the total number of pixels in the segmented region, and (x_i, y_i) were the coordinates of each pixel. The centroid (x_a, y_a) represented the actual stem soil-entry location, which was marked with a yellow plus sign in Fig. 9 and used by the robotic knife. This meticulous approach ensured the precise localization of the crop plant's stem entry into the soil, contributing to the accuracy of the overall analysis.

3.2. Evaluation metrics

3.2.1. Classification accuracy

The proportion of correctly classified celery plants and weeds out of the total number of images processed was calculated as:

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{Total Images}} \quad (6)$$

where True Positives (TP) referred to the number of celery plants correctly identified as celery, and True Negatives (TN) referred to the number of weeds correctly identified as weeds. The total number of images analyzed was the sum of TP, TN, False Positives (FP) and False Negatives (FN). False Positives (FP) indicated the number of weeds incorrectly classified as celery, and False Negatives (FN) indicated the number of celery plants incorrectly classified as weeds.

3.2.2. Precision

The proportion of true positive predictions (correctly classified celery plants) out of all instances predicted as celery plants was calculated as:

$$\text{Precision} = \frac{\text{TP} + \text{FP}}{\text{TP}} \quad (7)$$

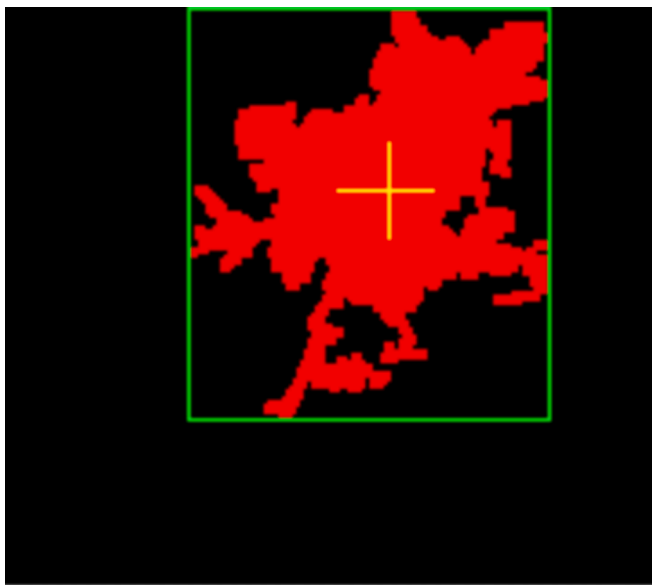


Fig. 9. Image represented weeds or background (black) and crop plants (red) with boundary box (green), center of mass (yellow plus) considered as stem soil-entry location.

3.2.3. Recall

The proportion of true positives (correctly classified celery plants) out of all actual celery plants in the dataset was computed as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8)$$

3.2.4. Soil entry point detection error

The Root Mean Square Error (RMSE) between the desired and actual stem soil-entry locations was computed as:

$$\text{RMSE} = \sqrt{(x_d - x_a)^2 + (y_d - y_a)^2} \quad (9)$$

where (x_d, y_d) represented the desired location and (x_a, y_a) represented the actual point of the stem soil-entry location. The mean and standard deviation errors for identifying the stem soil-entry location were calculated, respectively.

$$\text{Mean}(\bar{x}) = \frac{1}{n} \sum_{i=1}^n \sqrt{(x_d - x_a)^2 + (y_d - y_a)^2} \quad (10)$$

$$\text{Standarddeviation}(S_x) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (11)$$

4. Results and discussions

The results of this study demonstrate the efficacy of the proposed algorithm in accurately classifying celery crops and distinguishing them precisely from weeds under controlled experimental conditions. Key findings include the effectiveness of the photobleaching method in enhancing crop visibility for machine vision systems, leading to high crop detection accuracy and precise soil-entry point detection.

4.1. Weed and crop classification

The classification performance of the proposed algorithm was evaluated using 316 images captured from celery crop plants treated with the Rhodamine-B signaling compound, under standard commercial farming practices at the USDA research station in Salinas, CA. These images, with dimensions of 1608 x 1608 pixels, included both celery crops and weeds (mustard and dill) with varying densities, resulting in a 0 % false-negative detection rate.

The stem soil-entry location for each celery plant was also determined with high precision. The mean and standard deviation errors for stem soil-entry location were 3.58 mm and 8.52 mm, respectively, across the 313 images. The algorithm's performance was further assessed by measuring its precision in locating the stem's entry point into the soil, which yielded a 99.66 % precision rate, with an average

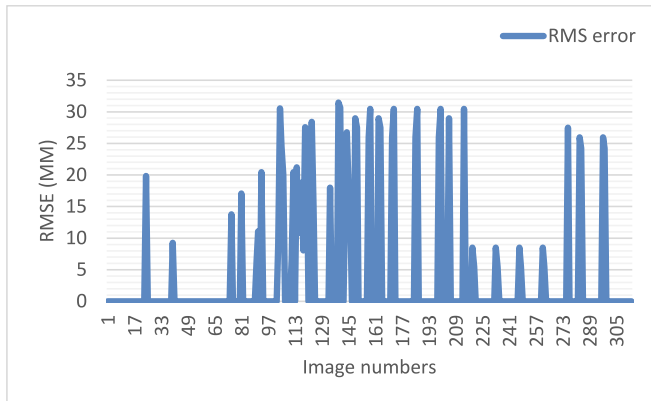


Fig. 10. RMS error between desired and actual stem soil-entry location with respect to the image number.

processing time of 30 ms per frame. Fig. 10 illustrates the accuracy of crop detection and the precision of stem soil-entry location for the analyzed images, confirming the algorithm's effectiveness in both classification and localization tasks. The system's performance is summarized in Table 1.

4.2. Comparative analysis

The comparative analysis of various studies focusing on weed detection and classification across different crops is summarized in Table 2. These studies utilized a range of sensor technologies, including RGB cameras, hyperspectral imaging, and machine vision systems, to tackle challenges posed by varying weed densities. The assessment considers factors such as the type of crop, detection methods, features employed, classification time, and accuracy metrics. While each study addressed specific challenges related to varying weed densities, crops, and environmental conditions, the proposed method demonstrates significant advancements in accuracy and speed, making it a more robust solution for precision agriculture. For instance, Lamm et al. (2002) and Wang et al. (2007) employed traditional machine vision systems, achieving moderate accuracies of 82.75 % and 70 %, respectively, reflecting the limitations of early machine vision techniques in crop-weed detection. Lamm et al. (2002) relied on size features, whereas Wang et al. (2007) used color indices in combination with GPS. Although effective under moderate weed densities, these approaches lacked the sophistication necessary for high-accuracy classification in complex environments.

In contrast, more recent studies, such as those by Knoll et al. (2019) and Jin et al. (2022), integrated deep learning and advance machine vision techniques, highlighting the technological progression in this field. For example, Knoll (2019) achieved an accuracy of 95.2 % using ResNet-50 with texture and color features, and Jin et al. (2022) attained 92.4 % accuracy leveraging CNNs and spatial-textural data. However, significant variations in processing speeds were observed, ranging from 10 ms per frame (Knoll, 2019) to a slower 8,333 ms (McCool et al., 2017) using DCNNs.

The proposed method, based on crop signaling, and leveraging photometric cameras, stands out by overcoming many limitations seen in earlier studies. While Raja et al. (2020a) successfully applied crop signaling in lettuce fields with accuracy of 98.93 %, the proposed method achieves a perfect classification accuracy of 100 %. This improvement is coupled with a significant reduction in processing time to 30 ms, compared to 160 ms in Raja et al. (2020c) and 50 ms in Milioto et al. (2017) for sugar beets.

The proposed method is simple, cost-effective, making it suitable for large-scale applications in resource-constrained environments. Its lower computational requirements and real-time capabilities make it particularly advantageous. However, inherent limitations, such as the reliance on hard-coded thresholds (e.g., for segmentation and morphological operations) and sensitivity to lighting variations, remain. These are areas where deep learning approaches, such as those by Su et al. (2021) and Zhang et al. (2022), are better equipped to address.

In summary, while previous studies have made significant progress in crop and weed detection and classification, many have faced trade-offs between speed and accuracy, or were limited by the specific crops and weed densities they targeted. The proposed method demonstrates a substantial improvement over these approaches, particularly in environments with high weed density. The proposed method represents a major advancement in accuracy, efficiency, and real-time functionality, especially in densely populated environments like celery fields. The novel and robust machine vision algorithm developed here is a promising solution for precision agriculture.

5. Conclusions

This study presents a novel approach within the crop signaling

Table 1

Analyzed results of the developed algorithm for weed-crop detection.

Number of images	Celery plants correctly classified	Celery classified as weeds	Weeds correctly classified	Weeds classified as celery	Stem soil-entry point detection accuracy	Processing time of each image
313	100 %	0 %	100 %	0 %	99.66 %	30 s f ⁻¹

Table 2

Comparison of various crop and weed classification technique.

Study		Sensors	Crop	Weed density	Detection and classification			
Category	References				Method	Features	Speed (ms f ⁻¹)	Accuracy (%)
Machine Vision-Based Approaches	Lamm et al. (2002)	RGB, 3-CCD scan camera	Cotton	Moderate	Machine vision	Size	160	82.75
	Wang et al. (2007)	Digital video camera	Wheat	Moderate	Machine vision, GPS	Colour indices	n/a	70
	Piron et al. (2011)	Black & white camera with filter wheel	Carrot	Moderate	Machine vision, 3D image information	Corrected plant height	n/a	83
Machine learning	Lin et al. (2017)	CCD camera (Model INFINITY3-1)	Corn	Low	Hyper spectral images & Decision trees	Combination of Spectral, Shape and Textural	n/a	95
	Zhang et al. (2022)	RGB Camera	Lettuce	Moderate	SVM with image blocking for overlap recognition	Color, texture, overlap features	Not reported	91.3
Deep learning	Zhao et al. (2022)	RGB + Hyperspectral Camera	Cabbage	Moderate	Machine learning with target spraying	Spectral reflectance features	Not reported	93.8
	Milioto et al. (2017)	RGB + NIR camera (JAI AD-130 GE)	Sugar beets	Low	CNNs	No hand-crafted features	50 ± 10 ms per blob	95
	McCool et al. (2017)	RGB camera	Carrot dataset, Haug and Ostermann (2014)	Low	DCNNs	Shape and statistical parameters	8333	93.9
	Jin et al. (2022)	RGB Camera	Vegetables	High	Deep Learning (CNN)	Spatial and texture	Not reported	92.4
	Knoll et al. (2019)	RGB Camera	Carrots	Moderate	Deep Learning (ResNet-50)	Texture and color features	10 ms	95.2
	Su et al. (2021)	RGB + Depth Cameras	Wheat, barley	Moderate	Deep Learning (CNN)	Semantic segmentation features	Not reported	94.5
Crop signaling and machine vision	Raja et al. (2020a)	RGB Basler 5 MP camera	Lettuce	High	Machine vision	Crop signaling	160	98.93
		Photometric camera (model: prime 95B 25 mm)	Celery	Very High	MachineVision	Crop signaling	30	100

framework by introducing Rhodamine B (Rh-B) as a fluorescent marker for precise identification of celery crops effectively distinguishing them from surrounding weeds. The crop signals on the plants are uniquely designed to have optical signatures that are machine-readable, which significantly simplifies and ensures the task of crop identification to discriminate from weeds. The system utilized computer vision techniques to successfully identified celery crop plants and determined the stem soil-entry location of each celery plant with 100 % accuracy with no false positives and a processing speed of 30 ms, demonstrating superior performance in distinguishing crops from weeds in high-density weed scenarios. The proposed algorithm outperforms many of the existing ones in terms of detection accuracy in critical in-field conditions including high weed densities.

While this study focuses exclusively on the correct identification of crops, its findings hold significant potential for improving robotic weed control by enabling targeted crop preservation in challenging agricultural settings. Future work will integrate the machine vision system with a knife control mechanism for testing in field, addressing variability in environmental conditions such as lighting, soil types, and weed density to validate its robustness and practical applicability. This research lays the foundation for advancing precision agriculture through efficient crop identification techniques.

CRediT authorship contribution statement

Rekha Raja: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Wen-hao Su:** Writing – original draft, Methodology. **David C. Slaughter:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Steven A. Fennimore:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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